Summer Break Holiday Homework 2026-27 Class 10

ENGLISH

MULTIPLE CHOICE QUESTIONS-

(From The Glove and the Lions, The Elevator, and Julius Caesar (Act 3 Scene 2,3).

1.Why did the knight throw the glove back at the lady?

A. He was afraid of lions

B. He wanted to insult the king

C. He valued self-respect over false love

D. He forgot the glove

2.What is the best real life lesson from Martin's fear in The Elevator ?A.Avoid elevators forever

B.fear should alway be ignored.

C.Gradually face and overcome irrational fears D.Trust only instincts without thinking

3. Why does Mark Antony call Brutus honourable repeatedly.?"

A. To praise him genuinely

B. To sarcastically influence the crowd against him. C.Because the admires Brutus

D. Because he is afraid of Brutus.

4. Why does the crowd turned violent in Act 3 Scene 3?

A. The understand politics clearly.

B. They are influenced by emotions.

C. They are trained soldiers.

D. They follow laws strictly.

5. What does the knights final action of throwing the glove back suggest?

A. Revenge is justified.

B. Blind love should always be followed.

C. Self respect is more important than fake admiration

D. Fear controls brewery.

Reason based questions:

6.Why did the lady throw the glove into the lion's den? What does this act reveal about her character?

7. Why do you think the king and the court were watching the lions? What does it tell us about their society?

8. Why does Martin's fear increase as the story progresses?

9. Why does the crowd turn against the conspirators?

10. Why does Mark Antony call Brutus "honourable "?

Competency based question:

11.What does the lady's behaviour teach us about pride and human nature?

12. What does the mob's reaction teach us about public thinking?

13. What do parents do when children face irrational fear like Martin?

14. What lesson does the story give about facing fear instead of avoiding them. How can this be applied in daily life?

15. What does the knight's final reaction tell us about self respect and dignity in relationship?

16. Was the lady's act of throwing the glove into the lions den a test of love or pride? Explain your viewpoint?

17. Do you think Martin's fear of the old woman is real or imagined?Justify your answer with the evidence from the story.

18. What does the killing of Cinna the poet reveal about how crowds behave when influenced by emotions?

19. What does the killing of Cinnathe poet suggests about mob psychology?

20. What does Brutus' trust in Antony reveal about his personality and political judgement?

Application based question:

21. In real life,people sometimes develop fear of certain places (like lifts, dark rooms or crowded spaces). How can Martin's experience help us understand and deal with such fears?

22. How does the behaviour of Martin's father influence Martin's fear? Do you think his approach was helpful or harmful? Give reasons.

23. How can the knights final action of throwing the glove back be applied in real life relationships where respect is not valued.

24. How should parents or elders help a child who is afraid of using an elevator instead of forcing them like Martin's father?

25. How can the knight’s rejection of the lady after retrieving the glove be applied in situations where people expect blind obedience?

26. How is the lady’s ‘challenge’similar to today's trends where people are daredto do risky task for entertainment or approval?

27. How can the persuasive techniques used by Mark Antony be applied by leaders or speakers today to influence people in a positive way?

28. If you were in Brutus position, how would you handle Antony’s request to speak at caesar's funeral differently?

29. How could Brutus have communicated his reasons for Caesar ‘s assassination more effectively to avoid misunderstanding?

30. How does the incident of Cinna- the poet highlight the importance of acting responsibly in groups or public spaces?

....

PROJECT WORK (ENGLISH LITERATURE PROJECT FILE)

<u>Poem-The Glove and the Lions</u>

•Short summary of the poem

•Character analysis of the lady and the knight

•Moral of the poem

<u>Short Story-The Elevator</u>

•Summary of the story

•Character sketch of Martin

<u>Drama-Julius Caesar</u>

•Summary of Brutus and Antony’s speeches

(Note: Word limit is for all the three topics in about 1000-1500 words.)

<u>Reminder-</u>

Project file will include-

1.Introduction

2.Certificate

3.Index

4.Acknowledgement

5.Content

6.Bibliography

(File cover should be black with silver taping.)

HINDI

1. निम्नलिखितमेंसेकौनसाशब्दपुल्लिंगहै?

(क) माता (ख) पिता (ग) नदी (घ) बहन

2. "लड़का" शब्दकाबहुवचनक्याहै?

(क) लड़के (ख) लड़कियाँ (ग) लड़का (घ) लड़कों

3. "सुंदर" शब्दकाविलोमक्याहै?

(क) बदसूरत (ख) सुंदरता (ग) सुंदराई (घ) सुन्दरतम

4. निम्नलिखितमेंसेकौनसावाक्यशुद्धहै?

(क) वहमुझेपैसेदेगा।

(ख) वहमुझपैसा देगा।

(ग) वहमुझेपैसेदेगा ।

(घ) वहमुझेपैसादेंगे।

5. निम्नलिखितमेंसेकौनसाशब्दविशेषणहै?

(क) लड़का (ख) सुंदर (ग) पढ़ना (घ) पुस्तक

6. प्रेमचंदकासंक्षिप्तजीवनपरिचयलिखिए।

7. सम्मिलित कुटुम्ब केसंबंधमेंश्रीकंठ सिंह केक्याविचारहैं?

8. बड़ीबेटीहोनेकेनातेपरिवारमेंबेटियोंकीक्याभूमिकाहोतीहै?

9. आनंदीऔरलाल बिहारी कीतकरारकिसबातपरशुरूहुई?

10. जयशंकरप्रसादकासंक्षिप्तजीवनपरिचयलिखिए।

11. वक्ता रामनिहाल अपनेआपकोचतुरक्योंसमझनेलगाथा?

12. मनोरमा औरश्यामादोनोंपात्रोंमेंक्याअंतरहै? कहानीकेआधारपरबताइए।

13. रामनिहाल कोकिससेप्रेमहोगयाथा?

14. ब्रजकिशोर बाबू कैसा व्यक्ति था ?

15. लीलाधर शर्मा पर्वतीय कासंक्षिप्तजीवनपरिचयलिखिए।

16. "भीड़मेंखोयाआदमी" कहानीकेशीर्षकसे आपक्यासमझतेहैं?

17. सूरदासकासंक्षिप्तजीवनपरिचयलिखिए।

18. यशोदामाँथपकीदेकरकिसकोसुलानेकाप्रयासकररहीहैं?

19.माखन खाते समय श्री कृष्ण कैसी-२ चेष्टांये कर रहे हैं ?

20. कृष्णभगवानक्यालेने कीजिदकररहेहैं? माँयशोदाउनकीजिदपूरीकरनेमेंक्योंअसमर्थहैं?

21. “खीजत जात माखन खात " इसपंक्ति काआशयस्पष्टकीजिए।

22. तुलसीदासकासंक्षिप्तजीवनपरिचयलिखिए।

22."विनय के पद" तुलसीदास की किस कृति से लिए गये हैं?

23."ऐसो को उदार जग माही" पंक्ति का आशय स्पष्ट कीजिए।

24.तुलसीदास किन्हें तथा क्यों त्यागने को कह रहे हैं?

25.'गीध' और 'सबरी' कौन थे? राम ने उन्हें कौन-सी गति प्रदान की?

26."अंग-अंग ढीला होना" मुहावरे के आधार पर अर्थ लिखते हुए वाक्य बनाइये।

27."आँखों का तारा" मुहावरे का अर्थ स्पष्ट करते हुये वाक्य प्रयोग कीजिए।

28."समय का महत्त्व" उद्‌घाटित करते हुए 100 से 200 शब्दों में एक कहानी लिखिए।

29.'संस्कृति' शब्द का उपयोग करते हुए एक प्रस्ताव लेखन कीजिए।

30. परियोजना कार्य- "पुस्तकें जीवन की कर्णधार हैं" - पुस्तकों की मानव जीवन में उपयोगिता व वर्तमान में घटते वर्चस्व को आधार मानकर उनके शाश्वत महत्व पर प्रकाश डालते हुए लगभग 1000 शब्दों में परियोजना तैयार कीजिए।

MATHS

1-The roots of the equation. $X^2 - 5x + 6 = 0$ are:

A) 2, 3B) -2, -3. C) 1, 6. D) 0, 6

2- The nature of roots of $x^2 + 4x + 5 = 0$ is:

A) Real and distinct. B) Real and equal. C) Complex (non-real). D) Rational

3 - If one root of $x^2 - 7x + k = 0$ is 3, then value of k is:

A) 10. B) 12. C) 14. D) 21

4 - The sum of roots of $2x^2 - 3x + 1 = 0$ is:

A) 3/2. B) -3/2. C) ½. D) -1/2

5 - The remainder when $x^3 - 2x^2 + x$ is divided by $(x - 1)$ is:

A) 0. B) -1. C) 1. D) none

6 - If $(x - 2)$ is a factor of $x^3 + ax^2 - x - 2$, then value of a is:

A) 1. B) -1. C) 2. D) -none

7 - Solve: $2x - 3 > 5$

A) $x > 4$B) $x < 4$. C) $x \geq 4$. D) $x \leq 4$

8 - solve : $-3x + 6 \leq 0$

A) $x \geq 2$. B) $x \leq 2$. C) $x \geq -2$. D) none

9- The solution set of $x + 2 < 5$ is:

A) $x < 3$. B) $x > 3$. C) $x \leq 3$. D) $x \geq 3$

Choose the correct option for each:

A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion

B) Both Assertion and Reason are true, but Reason is not the correct explanation

C) Assertion is true, but Reason is false

D) Assertion is false, but Reason is true

Q-10 Assertion (A): The equation $x^2 - 6x + 9 = 0$ has equal roots.

Reason ®: If discriminant $b^2 - 4ac = 0$ the roots are equal.

Q -11. Assertion (A): The sum of roots of $x^2 - 5x + 6 = 0$ is 5.

Reason ®: For $ax^2 + bx + c = 0$ sum of roots is $-b/a$

Q-12 Assertion (A): The roots of $x^2 + 4x + 8 = 0$ are real.

Reason ®: If discriminant is negative, roots are imaginary.

Q-13 Assertion (A): The remainder when $x^2 + 3x + 2$ is divided by $(x+1)$ is 0.

Reason ®: By Remainder Theorem, remainder = $f(-1)$

Q-14 - Assertion (A): If $(x-3)$ is a factor of a polynomial, then $f(3) = 1$

Reason ®: A factor gives zero remainder.

Q -15 Assertion (A): Solving $-2x > 6$ gives $x<3$

Reason ®: When multiplying or dividing an inequality by a negative number, the inequality sign changes.

Q-16 Assertion (A): The solution of $3x + 2 \leq 11$ is $x \leq 3$

Reason ®: Linear inequations are solved like linear equations.

Q-17 rectangular garden has length 5 m more than its breadth. If its area is 84 m^2, find its dimensions using a quadratic equation.

Q-18 The product of two consecutive positive integers is 132. Form a quadratic equation and find the integers.

Q-19 A ball is thrown upward with initial velocity. Its height after time t seconds is given by

$h = -5t^2 + 20t$. Find the time when the ball hits the ground.

Q-20 A taxi charges ₹ 50 as base fare plus ₹ 10 per km. If a person has at most ₹ 150, how far can they travel? Form and solve an inequation.

Q-21A student must score at least 40 marks to pass. If they already scored 28 marks in internal assessment, what minimum marks are required in the exam (out of 80)? Form an inequation.

Q-22. A sum of ₹ 10,000 is deposited in a bank at 5% per annum simple interest for 3 years. Calculate the interest and total amount.

Q-23. A person deposits ₹ 500 per month in a Recurring Deposit for 2 years at 6% p.a. Find the maturity amount (use RD formula).

Q-24 A company offers a salary of ₹ 5000 per month with an annual increment of ₹ 200. In how many years will the salary be at least ₹ 6000?

Q-25 car rental charges ₹ 50 per day and ₹ 10 per km. If a customer has a budget of ₹ 1500, what is the maximum distance they can travel in a day?

Q-26 student scores 70 marks in a test. The passing mark is 40. How many additional marks are needed to pass?

Q-27 A person saves ₹ 2000 pr month. How many months will it take to save at least ₹ 50,000?

Q-28 two consecutive natural numbers whose product is 56. Find numbers

Q-29'Solve : $\left(1 + \frac{1}{x+1}\right)\left(1 - \frac{1}{x-1}\right) = 7/8$

Q-30Find the Three consecutive largest positive integers such that the sum of one – third of the first , one – fourth of the second and One – fifth of the third is at most 20 .

Q-31 Find three smallest consecutive whole numbers such that the difference between one -fourth of the largest and one fifth of the smallest is at least 3.

Q-32 Find the value of m , if the following equation has equal roots:

$(m - 2)x^2 - ((5 + m)x + 16 = 0$

Q-33 Sum of the squares of two positive odd numbers is 74.Find the numbers

Q-34 Solve and graph the solution set of given inequation on the number line.

Q-35 $4x - 19 < 3x/5 - 2 \leq (-2/5) + x, \quad x \in R$

Q-36 Solve : $\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 = 0$ (3)

Q-37. motor-boat, whose speed is 9 km/h in still water, goes 12 km downstream and comes back in a total time of 3 hours. Find the speed of the stream.

Q-38 . ₹ 250 is divided equally among a certain number of children. If there were 25 children more, each would have received 50 paise less. Find the number of children.

Q-39 Solve : $a / (ax - 1) + b / (bx - 1) = a + b$, **where** $a + b \neq = 0$ **and** $ab \neq 0$ **.**

Q-40 By increasing the speed of a car by 10 km / hrs , the time of journey for a distance of 72 km is reduce by 36 minutes. Find the original speed of the car.

Q-41A rectangular garden has an area of 96 m². If the length is 4 m more than the breadth, find the dimensions of the garden.

Q-42. The product of two consecutive positive integers is 132. Find the integers.

Q-43 The area of a square is equal to the area of a rectangle with length 12 m and breadth x m. Find x.

Q-44 difference between the squares of two consecutive numbers is 15. Find the numbers.

Q-45 Use remainder theorem and find the remainder when the polynomial g(x) = $x^3 + x^2 - 2x + 1$ is divided by x – 3.

Q-46$x^3 + 3x^2 - kx + 4$ is divided by (x – 2), the remainder is k. Find the value of k.

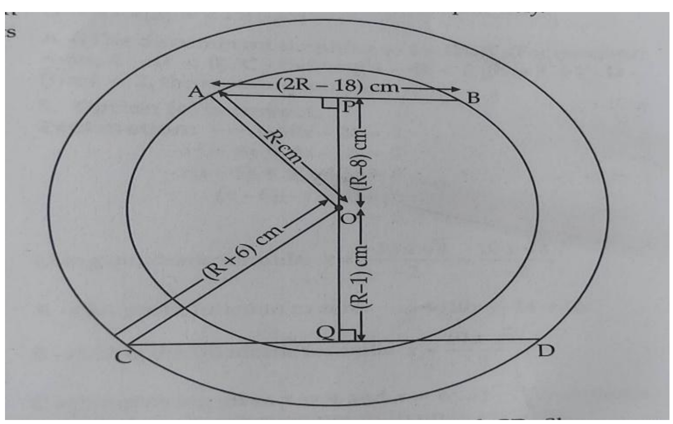
Q-47 Souhardya deposited ₹ . 800 per month for 1 year and received ₹ . 10,100 as the maturity value. Find the interest received by him .

Q-48. Solve : $\frac{1}{p}+\frac{1}{q}+\frac{1}{x}=\frac{1}{x+p+q}$

Q-49 By increasing the speed of a car by 10 km / hrs , the time of journey for a distance of 72 km is reduce by 36 minutes. Find the original speed of the car. (3)

Q-50 In the given figure below, two concentric circles have center O . Their radii are R and (R + 6) cm.respectively.

Find the length of the chord AB and CD ; show your working. (3)



Prepare a project on the following topics: (at least 40 pages)

(All to be included in a single black cover file)

1) Planning a home budget along with GST of articles/items purchased by you and your parents in the whole month.

2) Similarities of Triangles and other polygons.

3) 10 Indian + World Mathematician, their introduction, contributions and their achievements in the field of mathematics and science.

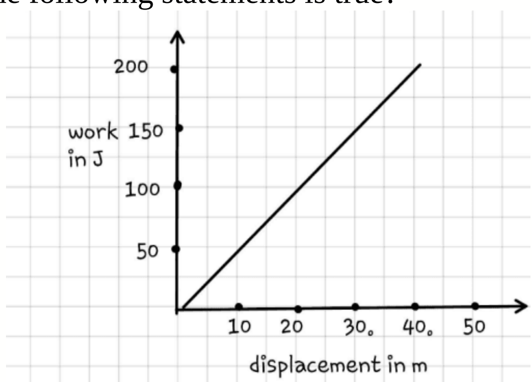
4) Survey of various types of Bank accounts, rates of interest offered.

5) Conduct a survey in your locality to study the mode of conveyance / Price of various essential commodities / favourite sports. Represent the data using a bar graph / histogram and estimate the mode

PHYSICS

Questions 1 to 10 each 1 marks

1) For producing the maximum turning effect on a body by a given force, the perpendicular distance of the line of action of force from the axis of rotation should be :
a) minimum
b) it does not matter
c) maximum
d) zero
2) A man moves in a circular path in a horizontal plane, the work done is :
a) Positive
b) Negative
c) Zero
d) Cannot say
3) The given figure depicts the graph of work done vs. displacement under a constant force of 10 N. Which of the following statements is true?



a) Force is acting at an angle of 0° with the displacement.
b) Force is acting at an angle of 45° with the displacement.
c) Force is acting at an angle of 60° with the displacement.
d) Force is acting at an angle of 90° with the displacement.

4) When a ball m is thrown upwards to a height h, the work done by the force of gravity is:
a) mgh
b) -mgh
c) mg
d) mgh^2

5) A body is acted upon by two unequal forces in opposite directions, but not in the same line. The effect is that:
a) the body will only have rotational motion
b) the body will only have translational motion
c) the body will have neither rotational motion nor translational motion
d) the body will have rotational as well as translational motion.

Assertion Reason Questions

Directions:

Each of the following questions consists of two statements: an **Assertion (A)** and a **Reason (R)**. Answer them by selecting the correct option:

(a) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

6) **Assertion (A):** When a body falls freely under gravity, mechanical energy is conserved.
Reason (R): The loss in potential energy equals the gain in kinetic energy.

7) **Assertion (A):** Work done by centripetal force on a particle in uniform circular motion is zero.
Reason (R): Work done is zero if force is perpendicular to displacement.

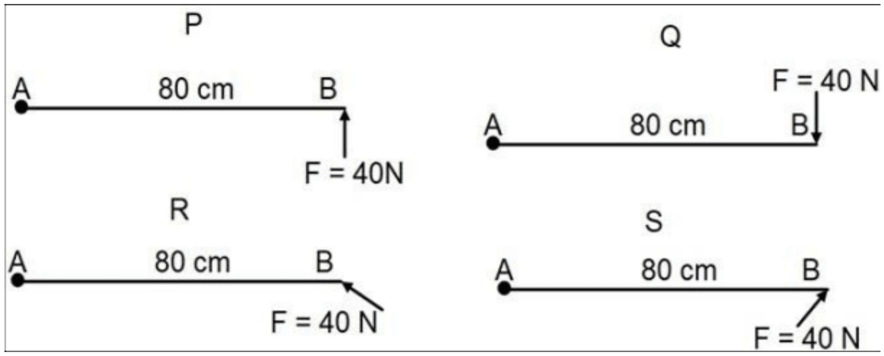
8) **Assertion (A):** A spring has minimum potential energy when it is compressed.
Reason (R): Potential energy stored in a spring is .$\mathbf{P} = \frac{1}{2}\mathbf{k}x^2$

9) **Assertion:** The turning effect on the body about an axis is due to the moment of force (or torque) applied on the body.
Reason: The SI unit of moment of force is N-m.

10) **Assertion:** Equilibrium means either the body is at rest (static equilibrium) or in uniform motion (dynamic equilibrium).
Reason: When the resultant force is zero or there is no unbalanced force, the body is said to be in equilibrium.

Questions 11 to 15 each 2Marks

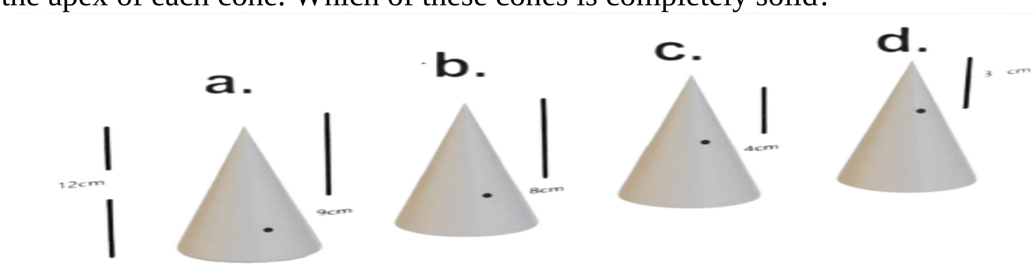
11)
The diagrams below show a force F = 40 N acting on a rod AB pivoted at A in different directions. Identify the correct statement.



1. P and S have opposite moments.
2. The magnitude of the moment of force is maximum in R.
3. The magnitude of the moment of force is maximum in P and Q.
4. The moment of force in R is negative.

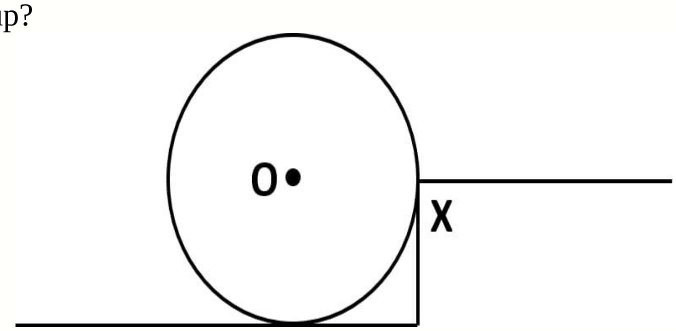
12) Soumya took two right circular cones of the same vertical height. One of the two cones is a solid one, while the other is hollow from inside. By measuring the cross-sectional areas of the cones and from the knowledge of symmetry, by using suitable formulas, he found the positions of the centre of gravity of both cones. He found that the difference is about 1.5 cm. What is the vertical height of the two cones

13) In the diagram below, four cones are depicted, each with a height of 12 cm. The position of the center of gravity is indicated by dots located at 9 cm, 8 cm, 4 cm, and 3 cm from the apex of each cone. Which of these cones is completely solid?



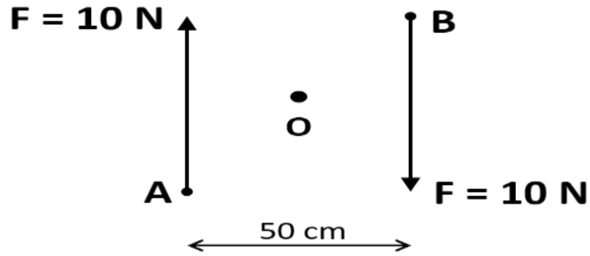
14) A 50 kg mass is being lifted by using a 1 metre long lever. If the length of the lever is reduced by 20%, how much more or less force is to be applied to achieve the same torque?

15) The adjacent diagram shows a heavy roller, with its axle at O, which is to be raised on a pavement XY. If there is friction between the roller and pavement, show by an arrow on the diagram the point of application and the direction of force to be applied. If pivoted at 0, now will it go up?



Questions 16 to 20 each 3Marks

16) Figure shows two forces each of magnitude 10N acting at the points A and B at a separation of 50 cm, in opposite directions. Calculate the resultant moment of the two forces about the point (i) A, (ii) B and (iii) O, situated exactly at the middle of the two forces.

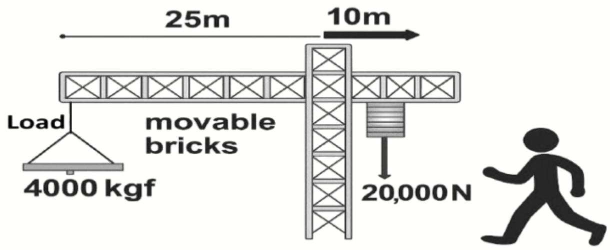


17) Regenerative braking involves harnessing the energy that is typically lost when a car decelerates and brakes and instead using it to recharge the car's batteries. Unlike traditional braking systems that simply dissipate energy, regenerative braking allows some of that energy to be recycled.
What is the energy conversion process that results in the wastage of energy in a normal car during braking?

18) How does regenerative braking differ from normal braking in terms of energy conversion?

19) Will the centre of gravity of a hollow sphere filled half with mercury and half with oil be identical to that of an empty hollow sphere? Give reasons for your answer.

20)
Study the diagram and answer the questions that follow.

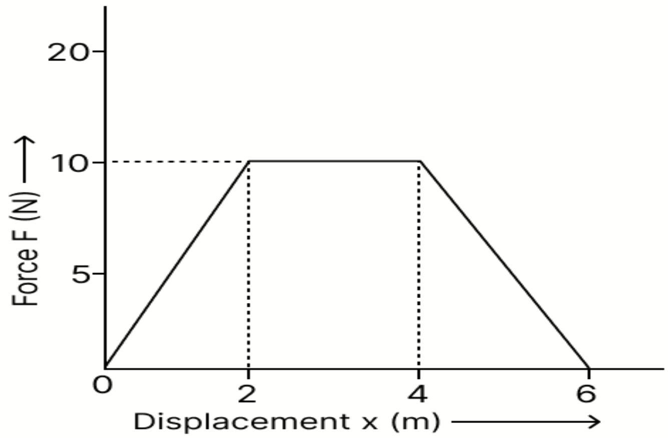


(a) In the diagram, is the worker attempting to raise or lower the load?

(b) Justify your answer to (a) with the necessary calculation.

Questions 21 to 25 each 2Marks

21) The graph given below shows a force-displacement graph. What is the total work done by force on the body in displacement from x = 0 to x = 6 m ?

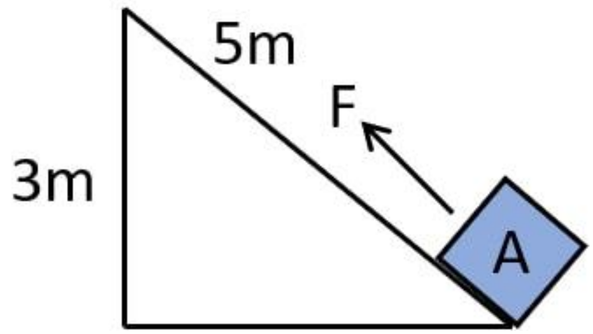


22) A coolie X carrying a load on his head climbs up a slope and another coolie Y carrying the identical load on his head move the same distance on a frictionless horizontal platform. Who does more work? Explain the reason.
23) What are the S.I. and C.G.S. units of work? How are they related? Establish the relationship
24) Define kilowatt hour. How is it related to joule?
25) State the condition when the work done by a force is (a) positive, (b) negative. Explain with the help of examples.

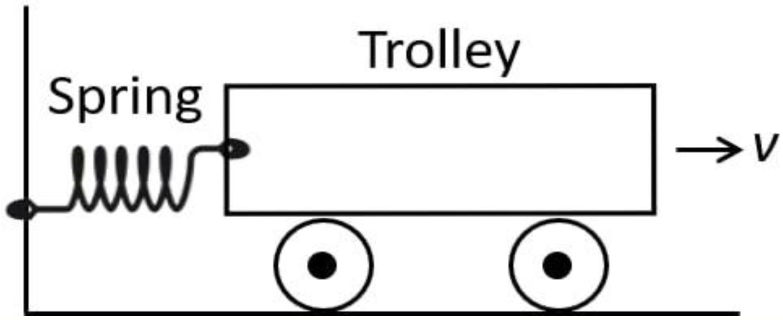
Questions 26 to 35 each 3Marks

26) An ox can apply a maximum force of 1000 N. It is taking part in a cart race and is able to pull the cart at a constant speed of 30 m s^{-1} while making its best effort. Calculate the power developed by the ox.
27) A body of mass 5 kg moves along a straight line under the action of a variable force. From x = 0 to x = 4 m, the force acts in the direction of displacement and increases linearly from 0 N to 20 N. From x = 4 m to x = 6 m, the motion of the body reverses while the force acts in forward direction. From x = 6 m to x = 10 m, the force becomes negative acting opposite to the original direction of motion. Calculate the net work done by the force on the body.

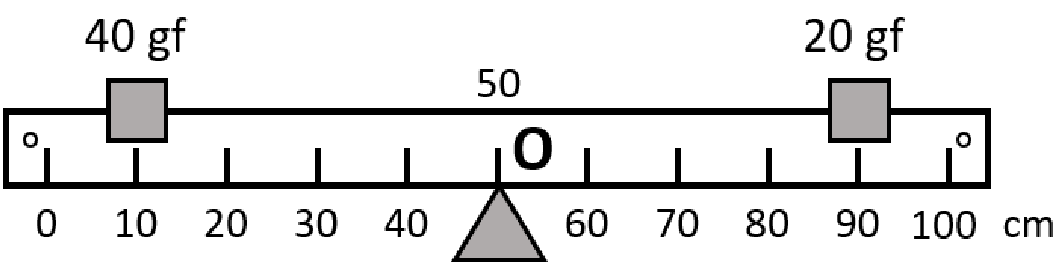
28) A body of mass m is moving with a uniform velocity u. A force is applied on the body due to which its velocity increases from u to v. How much work is being done by the force?
29) A block A, whose weight is 100N, is pulled up a slope of length 5 m by means of a constant force F (= 100 N) as illustrated.



30) A spring is kept compressed by a small trolley of mass 0.5 kg lying on a smooth horizontal surface as shown. When the trolley is released, it is found to move at a speed of $v = 2\ m\ s^{-1}$. What potential energy did the spring possess when compressed?



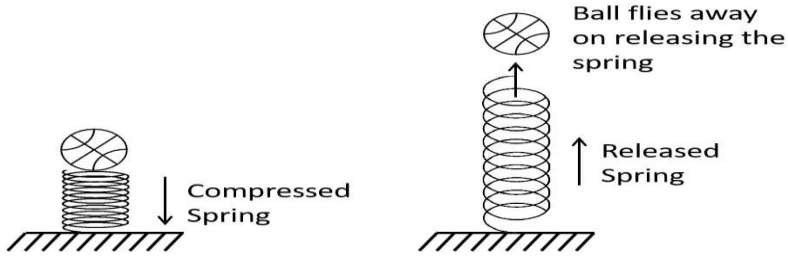
31) Figure shows a uniform metre rule placed on a fulcrum at its mid-point O and having a weight 40gf at the 10 cm mark and a weight of 20 gf at the 90 cm mark.
(a) Is the metre rule in equilibrium? If not, how will the rule turn?
(b) How can the rule be brought in equilibrium by using an additional weight of 40gf?



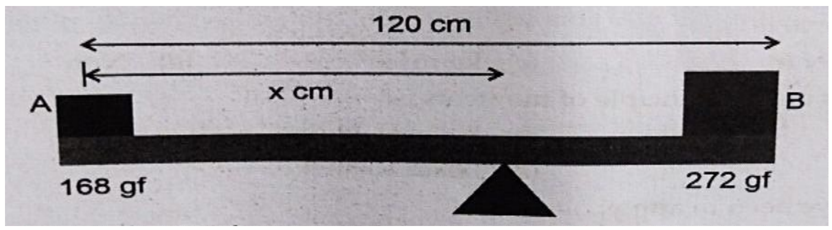
32) A uniform metre rule of mass 100 g is balanced on a fulcrum at mark 40 cm by suspending an unknown mass m at the mark 20 cm.
(i) Find the value of m.
(ii) To which side the rule will tilt if the mass m is moved to the mark 10 cm?
(iii) How can it be balanced by another mass 50 g?

33) (a) How does a centripetal force differ from a centrifugal force with reference to the direction in which they act?
(b) Is centrifugal force the force of reaction of the centripetal force?
(c) Compare the magnitudes of centripetal and centrifugal force.

34) A ball is placed on a compressed spring. What form of energy does the spring possess? On releasing the spring, the ball flies away. Give a reason.



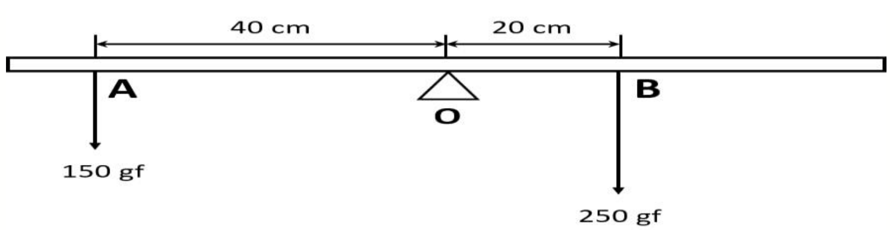
35) In the diagram, where A has a weight of 168 gf and B has a weight of 272 gf, and weightless beam is 120 cm long. Find the position of fulcrum from weight of 168 gf.



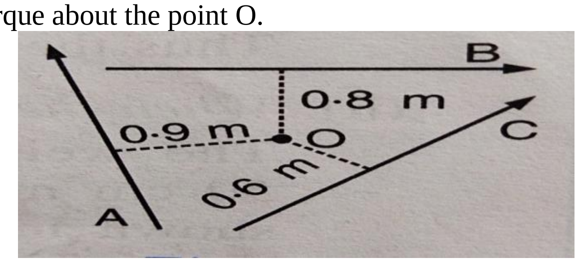
Questions 36 to 40 each 2Marks

36) A metre rod is half made of copper and half made of iron. If the mass of the copper part is 900 g and the mass of iron is 800 g, then calculate the position at which the rod can remain in equilibrium.

37) The diagram shows a uniform meter rule weighing 100 gf, pivoted at its centre O. Two weights 150gf and 250gf hang from the point A and B respectively of the metre rule such that OA = 40 cm and OB = 20 cm.Calculatethe distance from O where a 100gf weight should be placed to balance the metre rule.



38) A, B and C are three forces each of magnitude 4 N acting in the plane of paper as shown in fig. Point O lies in the same plane.
What is the resultant torque about the point O.



39) A simple pendulum while oscillating rises to a maximum vertical height of 7 cm from its rest position where it reaches its extreme position on one side. The mass of the bob of simple pendulum is 400 g and g = 10 m/s^2. Find the total energy of the simple pendulum at any instant while oscillating.

40) Name two examples in which the mechanical energy of a system remains constant.

CHEMISTRY

Multiple choice questions [1 to 5]

Q1. Which of the following is a weak acid?
a) HCl
b) H_2SO_4
c) CH_3COOH
d) HNO_3

Q2. An alkali is:
a) Insoluble base
b) Soluble base
c) Neutral salt
d) Acidic salt

Q3. The pH of pure water at 25°C is:
a) 0
b) 7
c) 14
d) 1

Q4. Which of the following is an acidic salt?
a) Na_2SO_4
b) NaCl
c) $NaHSO_4$
d) K_2SO_4

Q5. A base reacts with an acid to form:
a) Acid + Water
b) Salt + Water
c) Salt only
d) Water only

Assertion reason based questions [6 to 10]

Options:
A. Both Assertion and Reason are true, and Reason is the correct explanation
B. Both Assertion and Reason are true, but Reason is not the correct explanation
C. Assertion is true, Reason is false
D. Assertion is false, Reason is true

Q6.Assertion: Ammonium hydroxide is not used to test for lead ions.
Reason: Lead hydroxide is soluble in excess ammonium hydroxide.

Q7.Assertion: A white precipitate is formed when silver nitrate is added to sodium chloride solution.
Reason: Silver chloride is insoluble in water.

Q8. Assertion: Dilute hydrochloric acid is added before testing sulphate ions.
Reason: To remove interfering carbonate ions.

Q9. Assertion: Copper(II) ions give a deep blue solution with excess ammonium hydroxide.
Reason: Formation of a complex ion occurs.

Q10. Assertion: Barium chloride solution gives a white precipitate with sulphate ions.
Reason: Barium sulphate is soluble in dilute acids.

Competency / Application-Based Questions

Q11.A farmer’s field has low crop yield. Soil test shows low pH. What should the farmer do and why?

Q12. Tooth decay starts when pH of mouth falls below 5.5. Suggest preventive measures based on chemistry.

Q13.A student accidentally mixes acid with water improperly. What precaution should have been taken and why?

Q14. Why is it recommended to use antacids for acidity? Name a common compound used.

Q15.Why should curd and sour substances not be stored in metal containers?

Q16.Differentiate between strong acid and weak acid with one example each.

Q17.Why does dry HCl gas not change the color of dry litmus paper?

Q18.Write the reaction of zinc with dilute hydrochloric acid.

Q19.What is neutralization reaction? Give one example.

Q20.Explain why acids should be diluted before use.

□ **Chapter : Analytical Chemistry**

Q21.What is an indicator? Give one example.

Q22.Differentiate between qualitative and quantitative analysis.

Q23.What is titration?

Q24.Explain the role of indicators in acid-base titration.

Q25.Describe the steps involved in a simple acid-base titration experiment.

Q26.Explain how pH paper is used to determine the nature of a solution.

Q27.A solution turns phenolphthalein pink. What can you conclude about its nature? Explain.

Q28.During an experiment, a student gets inaccurate titration results. Suggest possible reasons.

Q29.How can you identify whether a given solution is acidic or basic without using pH paper?

Q30.A water sample is tested and found to have pH 6. What does this indicate about the water quality?

PROJECT WORK- Prepare display chart ,working and non working moldels regarding science exhibition.

Write the following experiment in lab manual

Recognition and identification of gases

- **Hydrogen chloride**
- **Hydrogen**
- **Sulphur dioxide**
- **Ammonia**
- **Hydrogen sulphide**

BIOLOGY

multiple choice questions

Q.1. Which of the following events occur during interphase?

P – DNA replication Q- Separation of chromatids

R- Protein synthesis S- Formation of spindle fibers

(a) P and R
(b) P, R and S
(c) All of them
(d) Q and S

Q.2. In plant cell division, which of the following are distinct features?

P – Presence of centrioles Q- Formation of cell plate

S- Constriction at the equator R – No asters

(a) Q and S
(b) P and R
(c) Q, R and S
(d) All of them

Q.3. How many genes does a child receive from its father?

(a) 25%
(b) 50%
(c) 75%
(d) 100%

Q.4. Law of independent assortment can be proved on the basis of which of the following ratios?

(a) 3:1
(b) 2:1:1
(c) 9:3:3:1
(d) 2:1

Q.5. Which of the following scenarios do not follow Mendelian inheritance?

P – A child has AB blood group from an A and B parent.

Q - A Career mother passes hemophilia to her son

R – A red and white flower produced pink offspring

S – A dwarf and tall pea plant produced only tall offspring

(a) P and Q
(b) Q and R

(c) R and S
(d) P and S

Section B assertion reason type questions

Each question consists of two statements namely assertion (A) and reason (R). For selecting the correct answer use the following code:

(a) Both assertion (A) and reason (R) are true and (R) is the correct explanation of (A)
(b) Both assertion (A) and reason (R) are true but (R) is not the correct explanation of (A)
(c) Assertion (A) is true and reason (R) is false
(d) Assertion (A) is false and reason (R) is true.

Q.1. Assertion (A): Mutations in DNA can lead to diseases like cancer

Reason (R): Mutations change the sequences of bases in DNA, which can alter the proteins produced.

Q.2. Assertion (A): Mitosis results in genetically identical daughter cells.

Reason (R): Mitosis equally distributes replicated chromosomes into two daughter cells.

Q.3. Assertion (A): If a mother is homozygous or black hair and father for blonde hair, their child can inherit black hair.

Reason (R): Gene for black hair is dominant to Jean for blonde hair.

Q.4. Assertion (A): Genes are specific sequences of bases in a DNA molecule

Reason (R): Each chromosome has only one gene.

Q.5. Assertion (A): Mitochondria can produce some of their own proteins.

Reason (R): Mitochondria contain their own DNA separate from the nuclear DNA.

Section C competency-based questions:

Q.1. A cell having 32 chromosomes undergoes mitotic division. During metaphase, what will the chromosome number (N) of the cell? During anaphase, what will the DNA content of the cell be?

Q.2. Under uncontrolled cell division, what is the pathological condition that occurs?

Q.3. Which cell between a eukaryote and a prokaryote has a shorter cell division time and why? Explain in detail.

Q.4. Write the phases of the cell cycle against each of the events

a) The disintegration of the nuclear membrane

b) The appearance of the nucleolus

c) Division of centromere

d) Replication of DNA

Q.5. How does cytokinesis in plant cells differ from that in animal cells?

Q.6. State Mendel's law of independent assortment.

Q.7.: How sex is determined in human beings (XY-type), and why the father determines the child's sex?

Q.8. Explain difference between Monohybrid and Dihybrid crosses.

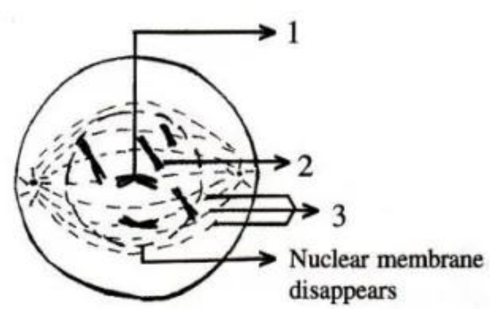
Q.9. Explain test cross.

Q.10. Write a short note on chromosome mutation and Down syndrome.

Section D diagram/Application based questions: -

Q.1.Draw a well labelled diagram of an animal cell with 4-chromosomes in metaphase of mitosis.

Q.2. Given below is a diagram representing a stage during mitotic cell division in an animal cell. Examine it carefully and answer the questions which follow.

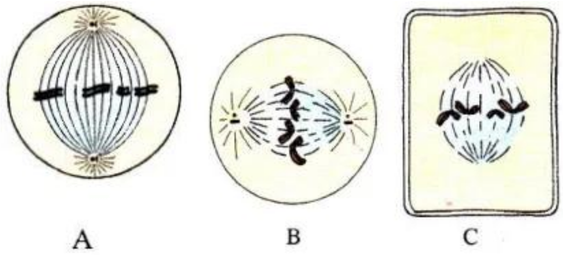


(a) Identify the stage. Give one reason in support of your answer.

(b) Name the cell organelle that forms the 'aster'

(c) Name the parts labelled 1, 2 and 3.

Q.3. Given below are three diagrammatic sketches (A, B and C) of one and the same particular phase during mitotic type of cell division.

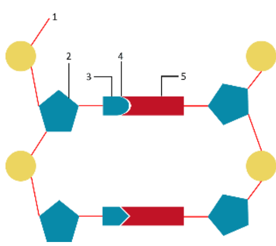


(a) Identify the phase.

(b) What is the diploid number of chromosomes shown in them?

(c) Identify whether A, B, C are animal or plant cells? Give reasons.

Q.4. Given below is a schematic diagram of a portion of DNA.



(a) How many strands are shown in the diagram?

(b) How many nucleotides have been shown in each strand?

(c) Name the parts numbered 1,2,3,4 and 5 respectively.

(d) Name the DNA unit constituted by the parts 1, 2 and 3 collectives.

Q.5. Given below are the sets of four terms. Choose the odd one and write the category of the remaining terms:

(a) Adenine, Guanine, Adrenaline, Thymine

(b) Pentose sugar, Histones, Phosphate group, Nitrogenous bases

(c) Metaphase, Anaphase, Interphase, Telophase

(d) G1 phase, M phase, G2 phase, S phase

(e) Chromoplast, Chromosome, Chloroplast, Leucoplasts.

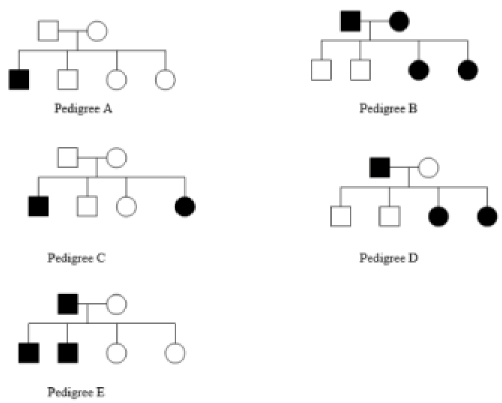
Q.6. Explain law of dominance with monohybrid cross.

Q.7. Explain law of independent assortment with Dihybrid cross.

Q.8. Write a short note on sickle cell anemia with diagram.

Q.9. Discuss the possible cases of color blindness (any two).

Q.10. Indicate the pattern of inheritance observed for pedigree A,B, C, D and E.



Section E project work :

Q.1. Prepare a project on anyone of the following :-

1. Write a project report on Mitosis (cell division) 10-15 pages

2. Prepare a explanatory chart on Mendel’s laws of Inheritance.

Q 2. Prepare a project on any one of the following:-

1.Monitoring plant growth.

Observe plants in your neighborhood and note their growth regularly.

2.Observation of birds.

Observe different types of birds in a park. Click photographs and make a collage.

3.Family history tree

Create a family tree showing geneticdisorders.

4.Leaf collection activity

Visit a park, collect different leaves, paste them in your notebook, and write the names of the plants.

HISTORY/CIVICS

Q1-Multiple Choice Questions

i- What is the strength of the Rajya Sabha for nominated members?

a) 12 b)13 c) 10 d) 15

ii- Arrange the following stages involved in the passing of a money bill.

i)The bill sent to the President for assent.

ii) Introducing of the bill in the Lok Sabha

iii) Rajya Sabha to return the billin 14 days.

iv) Bill sent to the Rajya Sabha.

a) ii,iv,iii,i b) i,ii,iii,iv c) iv,iii,ii,i d) iii,ii,i,iv

iii-The house is discussing an issue regarding the nation in the month of November.Which Session is in Motion?

a) Summer
b) Budget
c) Winter
d) Monsoon

iv-Who has the power to promulgate an ordinance?

a) Prime Minister
b) Vice President
c) Chief justice
d) President

v-JyotibaPhule has opposed the exploitation of Brahmin Priests in ………..

a) Neel Darpan
b) Gulamgiri
c) Amor Jiban
d) Bengal Gazette.

Q 2 Assertion and Reason based Questions

There are two statements given one labelled

Assertion(A) and the other labelled Reason

(R) select the correct answer to these questions from the code i,ii,iii iv are given blow

i) Both A and R are true and R is the correct explanation of the Assertion.

ii) Both A and R are true and R is not the correct explanation of the Assertion .

iii) A is true but R is false.

iv) A is False but R is true

a) Assertion- The Vernacular Press Act was called Gagging Act.
   Reason-It protested the Government with extensive rights to censor report and editorial of the Vernacular Press that would excite dissatisfaction against the Government.
b) Assertion-The Indian National Congress was founded by W C Banerjee
   Reason- The Objective was to end all racial religious and provincial prejudice and to promote a feeling of National unity among all the patriotism of the country.

c) Assertion- Although India had a long history going back to many centuries,it was never a single nation and comprised many kingdoms

Reason- The feeling of Nationalism emerged during the British rule mainly as a reaction to the British Rule

d) Assertion- Various political associations were formed in India by the the second half of the 19^{th} century to protect and promote general public interests.

Reason – The Indian National Congress was to first political association to be established in India.

e)) Assertion- The social reformers launched a crusade against socio-religious evils prevalent in Indian society in the 19^{th} century.

Reason – Many news papers and magazines in vernacular languages were launched.

Q-3 Extract Based Questions

He / She is the Constitutional head of the people of India under the Parliamentary system. The Executive Powers of the Union Government are exercised by him either directly or through officers subordinate to him .He/She is the head of the State. He /She is also the first citizen of India.

a) Who is he/She mentioned in the extract? What is the term of his/her office?
b) State any five Executive Power of he /She mentioned in the extract?
c) When he/She mentioned in the extract called the nominal head of Republic of India?
d) Mention any two discreationary powers of him.
e) How can a President remove from his office.

Q.4 a) if the Parliament has prorogued its session by the end of February ,with in which month it must have its next session? Why?

b) If Mr Ramlal want to become the President Which qualification to be need.

c) What is meant by the term Nationalism

d) Explain in brief about Illbert Bill.

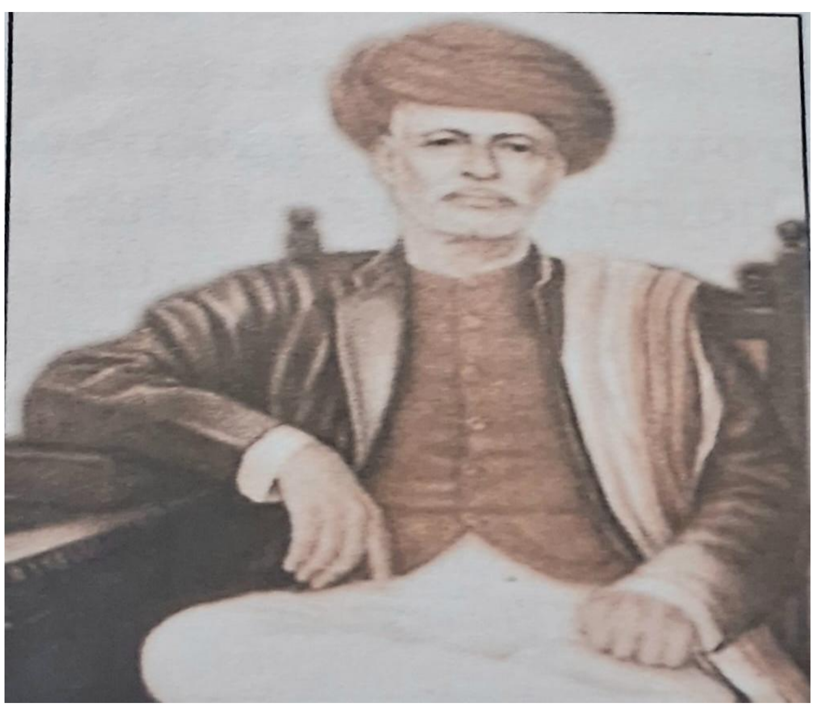
e) if Rohan want to study about Surendranath Banerjee, he should read about which Association with its objectives.

Q.5 a) picture study

i- Why it is called lower house of the Parliament?

ii- Write the composition of this house.

b)



i-Identify the famous social reformer.

ii- Which newspaper published by him?

ii-Write his two social contribution in the society.

c)

i- What is shown in this picture.

ii- How many membersdo President nominate to Rajya Sabha and who are they?

d)

Raja Ram Mohan Roy

i- Name the Association was established by him.

ii- Why he is called father of Indian Renaissance?

iii-Write any two contribution of him to reform Indian Society.

PROJECT WORK

Prepare a project file on any one of the given topic

1.Dr. Rajendra Prasad

2.Mahatma Gandhi- My experiment with truth

3. National Movement Period- 1917-1947

4.UN- organs

5-UN- Agencies

6- Non Aligned Movement

7- Malala Yusufzai or Kailash Satyarthi

8- Nehru’s Discovery of India

9- Abdul kalam’s Wings of fire

GEOGRAPHY

Q1Multiple Choice Questions

1.The recreating Monsoon season is experienced during…………

a) March – April. B) September – October

C) January – February. D) October – November

2.Which is the most widespread soil of India?

a)red soil b)black. C) laterite d) alluvial

3.Which region in India receives the highest annual rainfall?

a) Rajasthan. B) Thar desert. C) Punjab. D) Mawsynram

4.Which crop is most suitable for cultivation in black soil in India?

a)Wheat. B) cotton. C)rice. D) sugarcane

5) what causes snowfall in Kashmir during winter?

a) Tropical cyclone. B) North East

C) S.W monsoon. D) Temperate cyclone

Q2Assertion Questions

a) **Both (A) and (R) are true and ® is correct explanation of (A)**

B) Both (A) and (R) are true but ® does not expl**ain (A)**

C) (A) is true but (R) is false

d) (A) is false but (R) is true.

1.Asseration- The himalayasplay a crucial role in bringing heavy rainfall to India during summer season.

Reason – The himalayas act as a barrier that obstructed the rain bearing winds causing orographic rainfall on windward side

2. Assertion – Alluvial soil is very fertile and good for growing crops.

Reason – Alluvial soil is rich in potash and humus but poor in nitrogen and phosphorus.

3.Asseration- Soilero is a serious problem in the shivalik hills

Reason - Cutting down trees causes land slide and flood in this region.

4.Asseration - Black soil is called cotton soil.

Reason – Cotton grows well in black soil due to its moisture holding capacity.

5.Asseration- The northern plain are one of the most densely populated regions of India.

Reason –These plains are formed by hard igneous rocks and not suitable for agriculture.

Q3Competency based questions

2. The India Meteorological Department (IMD) reported that the El Niño phenomenon weakened this year, which could lead to favorable conditions for monsoon rainfall across India.

(*a*) Explain two ways in which the El Niño phenomenon affects the Indian monsoon.

(*b*) Name the wind system responsible for bringing moisture to the Indian subcontinent during the monsoon season.

2.Why is Rajasthan a desert even though the Southwest monsoon wind pass through it?

a) It receive rainfall only in winter.

b) The Aravali hills block the moisture before reaching Rajasthan

C) The Aravalli hills lies parallel to the direction of monsoon winds and low in in

Height .

In Kerala, Reena noticed that the soil on hilltops is soft, reddish in colour, and not good for farming without treatment. Her grandfather uses this soil to make bricks.

(*a*) What type of soil is being described?

(*b*) Give two characteristics of this soil. [*Apply*]

4.

Month	Jan	Feb	Mar	Apr	May	Jun
Temp (°C)	18.0	20.1	24.2	28.4	33.0	34.5
Rainfall (cm)	0.2	0.1	0.2	0.0	0.1	1.5

Month	Jul	Aug	Sep	Oct	Nov	Dec
Temp (°C)	34.0	33.2	31.1	27.3	22.5	19.3
Rainfall (cm)	10.4	14.5	12.0	1.3	0.5	0.2

(*a*) What is the driest month of the year? [*Understanding*]

(*b*) When does the rainfall sharply increase? [*Analysis*]

(*c*) Suggest the most likely reason for sudden rainfall rise in July and August. [*Evaluate*]

Deepak's farm is located in the Ganga plain. The soil i
fine, fertile, and good for growing wheat and sugarcane
However, during floods, fresh silt is deposited.

(*a*) Name the type of soil in Deepak's field.

(*b*) What makes this soil fertile every year? [*Appl*

6.

Priya notices that the land in her village in Madhya Pradesh has deep cracks, and small streams have cut channels into the fields. Farmers say they can't grow much there now.

(*a*) What kind of soil erosion is Priya observing?

(*b*) What can be done to stop this erosion? [*Apply*]

7.

1. Why does Kerala receive more rainfall than Tamil Nadu during the Southwest Monsoon season?
   (*a*) Kerala lies in the rain shadow region
   (*b*) Kerala faces the Bay of Bengal branch of the monsoon directly
   (*c*) Kerala lies on the windward side of the Western Ghats
   (*d*) Tamil Nadu receives rainfall only from the Arabian Sea branch [*Understanding & Application*]

A farmer in Tamil Nadu plans to grow rice during winter. Which climatic phenomenon should he rely on for irrigation? [*Application*]

(*a*) Western disturbances (*b*) Retreating Monsoon

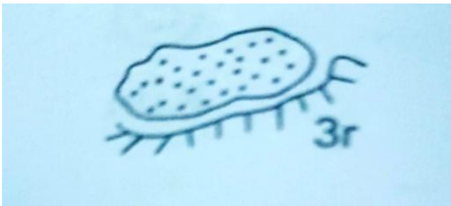
(*c*) Southwest Monsoon (*d*) Kal Baisakhi

9.Black soil has self plough ing property. Give reason.

10.Laterite soil is not suitable for cultivation. Give reason .

PICTURE BASED QUESTIONS

1.Identify the picture



2.

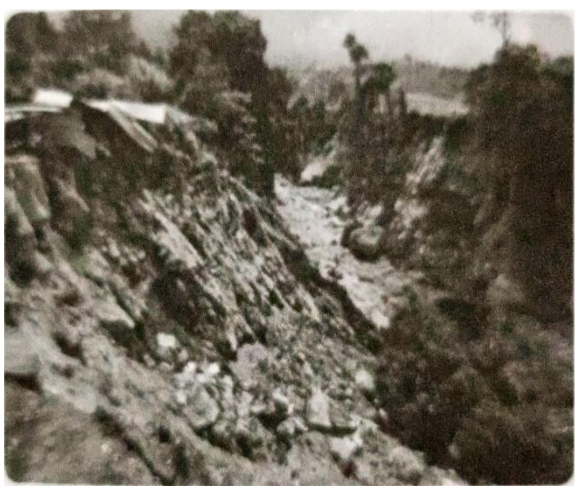
Identify and define

3.Name the agents responsible for soil erosion in areas shown in the picture

Identify the type of soil in the given map.

5.Looking at the image of cracked black soil suggest one crop that is best suited for the cultivation in this type of soil and explain why?

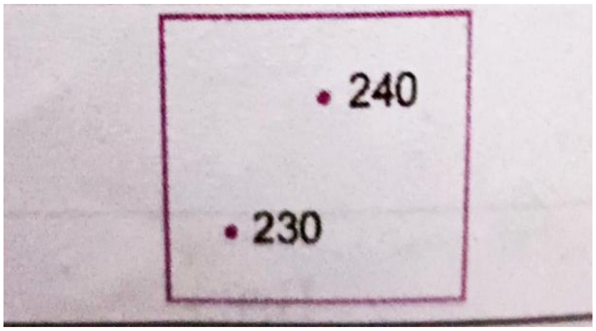
6.See the picture alongside which depicted an environmental disaster.



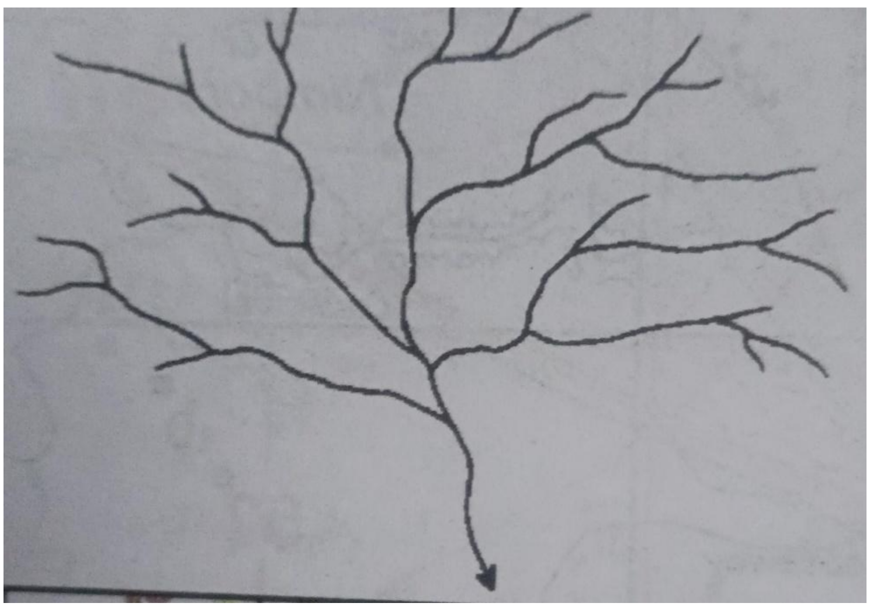
Identify

7.Define the term given in the picture

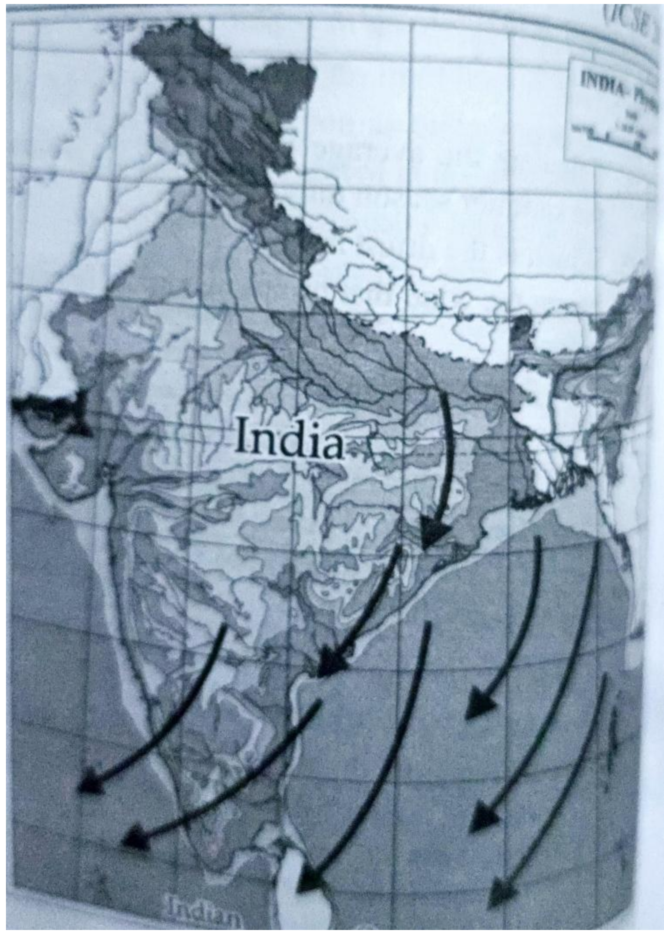
8.Identify the conventional sign



9.Idenfy the drainage pattern given in the picture.



10.Identify the wind in the given map



Project work

Make a project file on the topic "Transport in India". Cover with black chart paper with school logo.

COMPUTER

. Multiple Choice Questions :

1. **Which OOP principle allows a class to hide its data members from direct access by outside functions?**
   a) Polymorphism b) Inheritance c) Encapsulation d) Abstraction
2. **What is the default value of a boolean variable declared in a class?**
   a) true b) false c) null d) 0
3. **Which of the following is not a primitive data type in Java?**
   a) int b) char c) boolean d) String
4. **What is the result of the expression 15 % 4?**
   a) 3 b) 1 c) 3.75 d) 4
5. **What does Math.floor(-4.7) return?**
   a) -4.0 b) -5.0 c) 4.0 d) 5.0

II. Assertion and Reason Based :

Options for all questions:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

1. Assertion (A): In Java, int x = 5 / 2; stores 2.5 in x.
Reason (R): Division of two integers always produces an integer in Java.

2. Assertion (A): Math.random() returns a value between 0.0 and 1.0.
Reason (R): The return type of Math.random() is int.

3. Assertion (A): a = a + 5 and a += 5 are identical in performance.
Reason (R): += is a shorthand assignment operator.

4. Assertion (A): The final keyword allows a variable to be modified later.
Reason (R): final defines a constant in Java.

5. Assertion (A): System.out.println(Math.abs(-5)); prints -5.
Reason (R): Math.abs() returns the absolute value of a number.

III. Competency based Questions :

1. Why is Java called an Object-Oriented Programming language?
2. If you change a variable from private to public, how does it affect encapsulation?
3. What happens if you try to divide a number by zero in Java?

4. Why can't we use static methods to access non-static variables directly?
5. Describe the difference between Implicit and Explicit type casting with examples.
6. Why is String considered a class, not a primitive data type?
7. What is the output of System.out.println('a' + 'b');? Explain why.
8. How does the new operator create a connection between a class and an object?
9. What is the name of a member function that has the same name as the class and is used to initialize an object which does not have a return type.
10. Differentiate between next() and nextLine() in Scanner class.

IV. Application and Diagram based :

1. A software developer creates a system where a single "draw" function behaves differently depending on whether it is called by a "Circle" object or a "Square" object. Name the principle of Object-Oriented Programming (OOP) being applied here.

2. In a library management system, "Book" is defined with properties like title and author. A specific copy titled "The Alchemist" is added to the shelf. Identify which entity is the 'Class' and which is the 'Object'. Why is the object called an "instance of a class"?

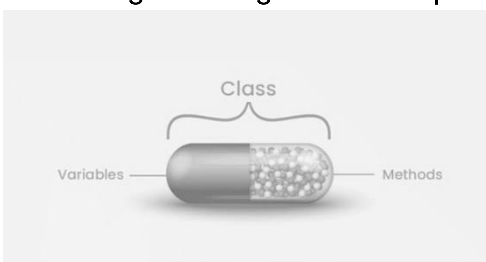
3. A student uses an int data type to store the value of a bank interest rate (e.g., 7.5%). Explain the error in this choice and suggest the correct data type.

4. Write a Java code snippet using the Math library to find the absolute difference between the squares of two numbers a and b.

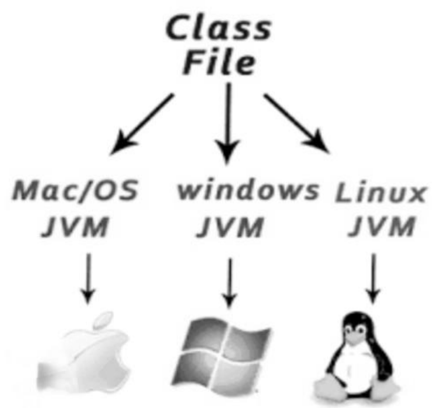
5. Identify the error in the following input segment:

```
import java.util.*;
class InputTest
 {
   public static void main(String args[])
    {
      Scanner sc = new Scanner(System.in);
      System.out.println("Enter age:");
      int age = sc.nextLine();
   }
}
```

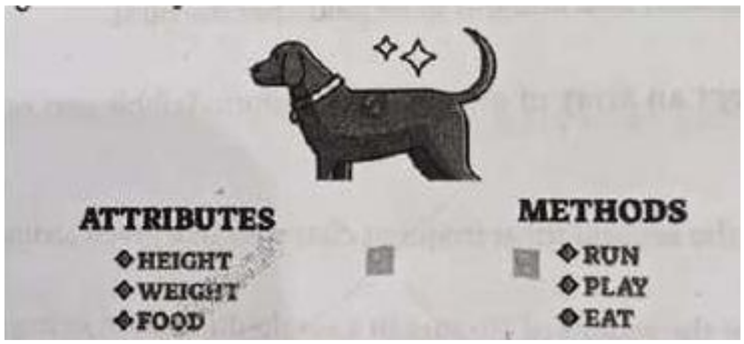
6. Name the OOPs principle for the given diagram and explain it.



7. Identify the object feature :

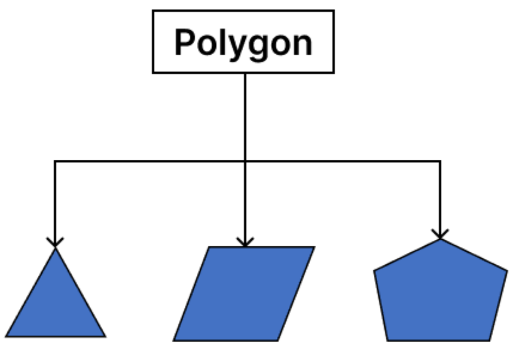


8. Which Oops principle shown in following picture

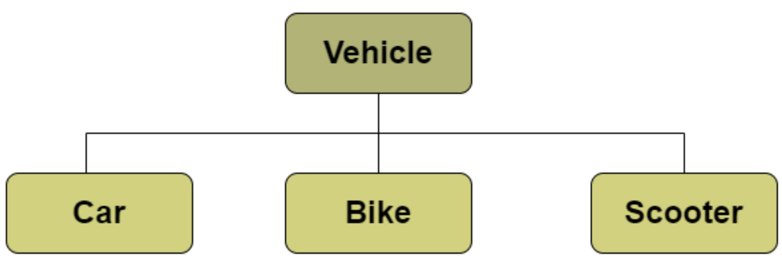


9. Which of following is a correct answer for given picture :

i.) Inheritance   ii.) Polymorphism   iii.) Absraction   iv.) Encapsulation



10. Differentiate and name the Class and Objects in the following picture.



PHYSICAL EDUCATION

MULTIPLE CHOICE QUESTIONS:

1) The stage from the age of 5 to12 years is called

a) infancy b)prenatal stage

c) adulthood d)childhood

2) The stage from the birth to end of 5 years of age is called

a) adulthood stage b)adolescence

c) infancy stage d)childhood stage

3) The ability of an infant to grasp an object is an example of

a)growth b development

c)recreation d)obesity

4)The aim of physical education is

a) Physical, social and mental development of a person

b)to develop co-ordination of body parts

c)to develop leadership

d) camadeirie

5)The process by which a child learns to interact with others around them is called as

a)physical development b)psychological development

c)emotional development d)social development

SHORT ANSWER QUESTIONS

1)What do you understand by the term growth?

2)What do you understand by the term development?

3)What do you mean by physical education?

4)Explain the term physical development?

5)Explain the term social development ?

LONG ANSWER QUESTION

1)Describe the infancy stage of development?

2) Describe the childhood stage of development?

3)Describe the adolescence stage of development?

4)Describe the adulthood stage of development?

5)State any three emotional development objectives ofphysical education

CASE STUDY QUESTION

1)Piya comes from a family with a history of obesity .Despite maintaining a healthy diet and exercise routine,she struggles to manage her weight .

In tne above case study name the factor which influence Piya's growt and development.

2)Jubin ,a 12 year old boy ,struggles tomake friends at school and often feels left out during physical activities.

Which of the following Development is concerned in Jubins case?

a)physical development b)social development

c)psychological development d)mental development

3)Simon, a 50 year old man ,has recently been diagnosed with hypertension and he find difficult to do daily routines activities

What will be the doctor's suggestion in this case?

a)take rest

b)participate in social activities

c)participate in physical education activities

d)watch television and relax

PICTURE STUDY QUESTIONS

1)Identify the following stage of growth and development.

COMMERCIAL STUDIES

MCQs

1. Which of the following is not a stakeholder?

(a) Customers

(b) Employees

(c) Machines

(d) Investors

2. Marketing focuses on:

(a) Production

(b) Customer satisfaction

(c) Accounting

(d) Storage

3. Services are:

(a) Tangible

(b) Perishable

(c) Storable

(d) Visible

4. Which pricing objective focuses on gaining market share?

(a) Survival

(b) Profit maximization

(c) Market penetration

(d) Cost coverage

5. Which is a sales promotion technique?

(a) Advertisement

(b) Personal selling

(c) Discounts

(d) Branding

Section B: Assertion & Reasoning

6. A: Stakeholders influence business decisions.

R: Businesses depend on stakeholders for resources.

Options:

A. Both A and R are true and R explains A

B. Both true but R not explanation

C. A true, R false

D. A false, R true

7. A: Services cannot be stored.

R: Services are intangible.

Options:

A. Both A and R are true and R explains A

B. Both true but R not explanation

C. A true, R false

D. A false, R true

8. A: Pricing affects demand.

R: Higher prices always increase demand.

A. Both A and R are true and R explains A

B. Both true but R not explanation

C. A true, R false

D. A false, R true

9. A: Sales promotion boosts short-term sales.

R: It attracts customers through incentives.

Options:

A. Both A and R are true and R explains A

B. Both true but R not explanation

C. A true, R false

D. A false, R true

10. A: Marketing is broader than selling.

R: Selling is only a part of marketing.

A. Both A and R are true and R explains A

B. Both true but R not explanation

C. A true, R false

D. A false, R true

Section C: Competency-Based Questions

11. Who are stakeholders?

12. Name any four types of stakeholders.

13. State expectations of customers from a business.

14. Differentiate between marketing and selling.

15. Distinguish between product and service.

16. What is pricing?

17. State any two objectives of pricing.

18. What is sales promotion?

19. Explain any two techniques of sales promotion.

20. What are services? Give examples.

21. Explain expectations of employees as stakeholders.

22. State two features of marketing.

23. What is the role of pricing in business?

Section D: Application-Based Questions

24. A company ignores customer complaints.

Which stakeholder is affected and how?

25. A firm reduces price to attract more customers.

Identify the pricing objective.

26. A salon provides haircuts and spa services.

Identify whether it deals in products or services and explain.

Section F: Real-Life & Image-Based Questions

27. (a) Identify the stakeholder shown

(b) State two expectations of this stakeholder

(c) Why are they important for business?

28. (a) Identify the marketing activity

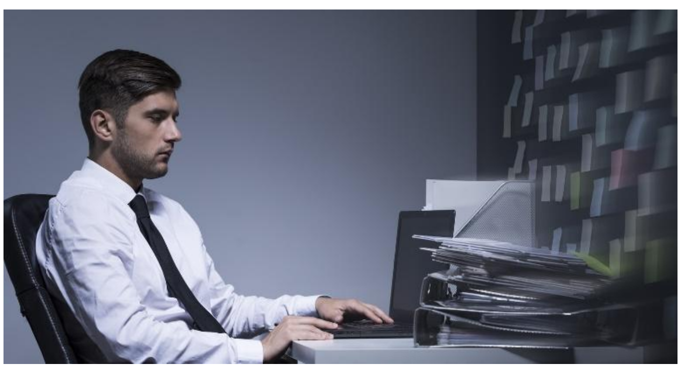
(b) Is it marketing or sales?

(c) State one benefit

29. (a) Product or service?

(b) Give two characteristics

(c) Why customer experience matters?



30. (a) Identify the stakeholder

(b) State two expectations

(c) Why satisfaction is important?

ECONOMICS

MCQs

(1)A fall in the price of a substitute good will:

a) Increase demand

b) Decrease demand

c) Not affect demand

d) Increase supply

(2)If demand is perfectly inelastic, the elasticity coefficient is:

a) Greater than 1

b) Equal to 1

c) Zero

d) Infinite

(3)Which situation shows contraction of demand?

a) Fall in income

b) Rise in price of the good

c) Increase in population

d) Change in taste

(4)Which factor causes a shift in supply curve?

a) Change in price

b) Change in technology

c) Movement along curve

d) Law of demand

(5)Cost-push inflation is caused by:

a) Increase in demand

b) Rise in production cost

c) Increase in supply

d) Decrease in taxes

 Section B: Assertion–Reason

(1)Assertion: Demand for inferior goods decreases when income rises.

Reason: Consumers shift to better quality goods.

(2)Assertion: Elasticity of demand is greater for luxury goods.

Reason: Luxury goods have many substitutes.

(3)Assertion: Supply curve shifts right due to technological improvement.

Reason: Production becomes more efficient and cheaper.

(4)Assertion: Inflation benefits debtors.

Reason: They repay loans with money of lower purchasing power.

(5)Assertion: Price elasticity is unitary when percentage change in demand equals price change.

Reason: Total expenditure remains constant.

 Section C: Competency-Based Questions

(1)A consumer buys more tea when coffee prices increase.

(a)Identify the concept involved.

(b)Explain the relationship between the two goods.

(2)Despite a sharp rise in petrol prices, its demand does not fall significantly.

(a)Identify the type of elasticity.

(b)Justify your answer with reasoning.

(3)A firm adopts new machinery and increases output even at the same price.

(a)What happens to the supply curve?

(b)Explain with reason.

(4)During inflation, fixed-income earners suffer more than businessmen.

(a)Explain why.

(b)Support with an example.

(5)When price falls from 50 to 40, demand increases from 100 units to 140 units.

Is demand elastic or inelastic?

Give reason (no formula required).

(6)A family reduces consumption of apples when prices rise.

(a)Identify the law applicable.

(b)Explain with reason.

(7)A shopkeeper notices that even after increasing price, sales of salt remain same.

(a)What type of demand is this?

(b)Why?

(8)During a festive season, supply of sweets increases.

(a)Explain why producers increase supply.

(9)Government increases taxes, leading to rise in prices.

(a)What economic condition is this?

(b)How does it affect consumers?

(10)A person shifts from eating out to home-cooked meals due to price rise.

(a)Which concept is shown?

(b)Explain briefly.

□□ Section D: Case Study / Application-Based (3×5 = 15 Marks)

Case Study 1: Demand Curve Analysis

A downward sloping demand curve shows that when price decreases from 20 to 10, quantity demanded increases.

Questions:

Explain the law illustrated.

Distinguish between movement along the curve and shift in demand.

Give one real-life example.

Case Study 2: Inflation Scenario

A report shows continuous rise in food prices due to increased fuel costs and supply disruptions.

Questions:

Identify the type of inflation.

Explain two causes based on the case.

Suggest one policy measure to control it.

Case Study 3: Supply Curve Situation

A graph shows supply increasing as price rises, but suddenly shifts right due to subsidy.

Questions:

Explain movement vs shift in supply.

What role does subsidy play?

Predict effect on market price.

Case Study 4: Elasticity in Real Life

A company reduces price of its product by 10% and notices a 25% increase in demand.

Questions:

Calculate nature of elasticity (no formula, only concept).

What type of good is this likely to be?

Suggest pricing strategy.

Case Study 5: Newspaper Extract

"Government announces measures to control rising inflation, including interest rate hike and reduction in public expenditure."

Questions:

Explain how interest rate affects inflation.

How does reduced government spending help?

Identify one limitation of such measures.

Case Study 6

A graph shows a downward sloping demand curve.

Questions:

Identify the curve.

State the law represented.

What happens when price falls?

Case Study 7

A chart shows prices of vegetables increasing continuously over months.

Questions:

What economic problem is shown?

Name two causes.

Suggest one remedy.

Case Study 8

A diagram shows an upward sloping supply curve.

Questions:

Identify the curve.

Why does it slope upward?

What happens when price rises?

Case Study 9

A picture shows people buying less petrol despite price increase.

Questions:

Identify the type of demand elasticity.

Explain the reason.

Give one similar example.

Case Study 10

A newspaper headline reads: “Inflation at 8% hits common people”

Questions:

What is inflation?

Who is most affected?

Suggest one control measure.